EXAMINATIONS COUNCIL OF ESWATINI
Eswatini General Certificate of Secondary Education

CANDIDATE NAME	

CENTRE NUMBER							CANDIDATE NUMBER					

MATHEMATICS **6880/01**

Paper 1 Short-Answer Questions (Core) **October/November 2025**

Candidates answer on the Question Paper. **1 hour**

Additional Materials: Scientific calculator
Geometrical Instruments

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name in the spaces provided.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do **not** use staples, paper clips, highlighters, glue or correction fluid.
Do **not** write on the bar code.

Answer **all** questions.

If working is needed for any question it must be shown below that question.
The number of marks is given in brackets [ ] at the end of each question or part question.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures.
Give answers in degrees to one decimal place.

For π, use either your calculator value or 3.142.

The total of the marks for this paper is 60.

For Examiner's Use	
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Total	

This document consists of **10** printed pages and **2** blank pages.

For Examiner's Use

1 **(a)** Write $\frac{132}{64}$ as a

(i) mixed number,

Answer (a)(i) .... [1]

(ii) percentage.

Answer (a)(ii) .... [1]

(b) Work out 3.157^2 correct to 2 decimal places.

Answer (b) .... [2]

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2 **(a)** Find the highest common factor of 72 and 48.

Answer (a) .... [2]

(b) Simplify the ratios

(i) 72 : 48

Answer (b)(i) .... [1]

(ii) 7 m : 42 cm

Answer (b)(ii) .... [2]

(c) The length and width of a rectangle are in the ratio, length : width = 3 : 2.

The width of the rectangle is 8 cm.

Find the length of the rectangle.

Answer (c) .... [2]

For Examiner's Use

3 $\underline{\boldsymbol{a}} = \begin{pmatrix} 4 \\ -8 \end{pmatrix}$, $2\underline{\boldsymbol{b}} = \begin{pmatrix} -6 \\ 12 \end{pmatrix}$ and $\underline{\boldsymbol{a}} - \underline{\boldsymbol{c}} = \begin{pmatrix} 3 \\ -10 \end{pmatrix}$

Find

(a) $\underline{\boldsymbol{b}}$,

Answer (a) .... [1]

(b) $\underline{\boldsymbol{c}}$,

Answer (b) .... [2]

(c) the magnitude of $\underline{\boldsymbol{a}}$.

Answer (c)..[2]

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4 Jojo borrowed E500 from a friend.

The friend charged him 10% simple interest per month.

He pays back a total of 850 after n months.

Find the value of n

Answer n = ...[2]

For Examiner's Use

5 **(a)** **(i)** Write 216 as product of its prime factors.

Write your answer in index form.

Answer (a)(i) .... [2]

(ii) Evaluate $16^{\frac{3}{2}}$ without using a calculator.

Show all your working

Answer (a)(ii) .... [2]

(b) Find the value of

$$\left(\frac{-14x^2y}{7xy}\right)^0$$

Answer (b) .... [1]

6 You are given three rows from Pascal's triangle.

In the space provided write the next row of Pascal's triangle.

1 4 6 4 1

1 5 10 10 5 1

1 6 15 20 15 6 1

.... [2]

For Examiner's Use

7 **(a)** A polygon has 8 sides.

Write down the mathematical name for this polygon.

Answer (a) .... [1]

(b) The interior angle of a regular polygon is 140°.

Find the number of sides of the polygon.

Answer (b) .... [2]

(c) The side of an equilateral triangle is 4 cm.

Construct the equilateral triangle.

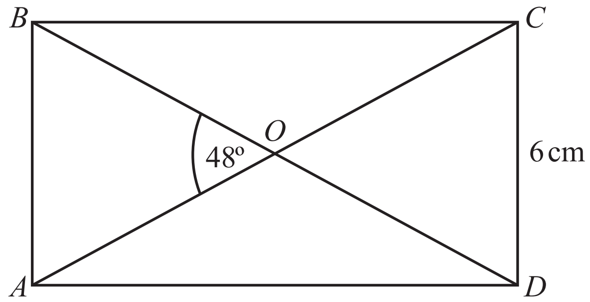
[3]

8 $ABCD$ is a rectangle.

Diagonals AC and BD meet at O.

$CD = 6$ cm.

Angle $AOB = 48°$.



NOT TO SCALE

(a) Find

(i) Angle CBO,

Answer (a)(i) .... [1]

(ii) Angle OCD.

Answer (a)(ii) .... [1]

(b) The area of triangle AOD is $12\,\text{cm}^2$.

Find the length of side AD.

Answer (b) .... cm [2]

---

9 Write down the name of a quadrilateral that has

(a) order of rotational symmetry 2 and no line of symmetry,

Answer (a) .... [1]

(b) order of rotational symmetry 2 and 2 lines of symmetry.

Answer (b) .... [1]

For Examiner's Use

10 One summer day, rain started falling at 1925 hours.

It stopped at 0423 hours the following day.

Calculate how long the rainfall lasted.

Answer .... hours .... minutes [2]

---

11 $f(x) = 2x - 1$

Find

(a) $f(7)$

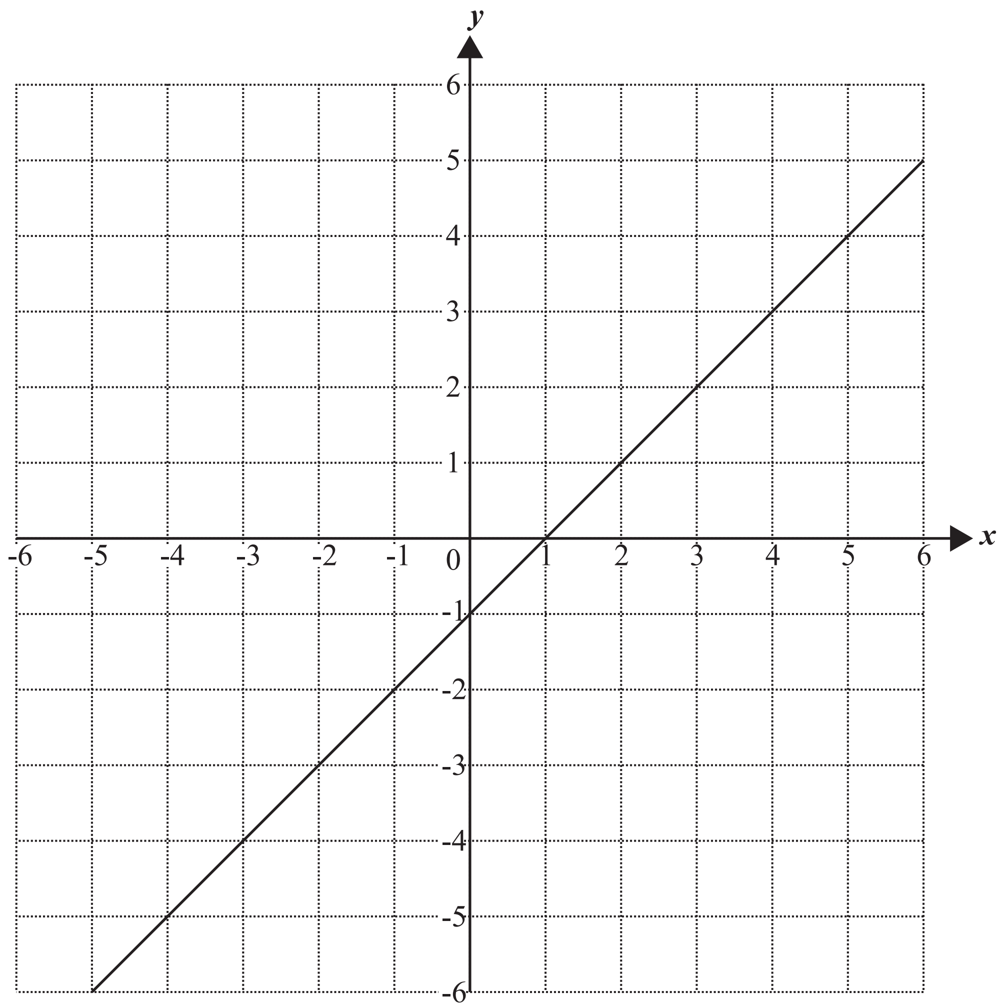
Answer (a) .... [1]

(b) $f^{-1}(x)$.

Answer (b) $f^{-1}(x) =$ .... [2]

For Examiner's Use

12 The grid shows a straight line.



(a) Find the equation of the straight line.

Answer (a) .... [2]

(b) A line has an equation $y = 3x - 5$.

State the gradient of the line.

Answer (b) .... [1]

---

13 Solve.

$2x + 5 = 29$

Answer $x =$ .... [2]

---

14 Expand and simplify.

$3(2x + 5) - 4(x - 2)$

Answer .... [2]

---

15 Solve the simultaneous equations.

$y = 3x$
$y = x - 8$

Answer $x =$ .... $y =$ .... [3]

---

16 Factorise.

$2px - 6py$

Answer .... [2]

---

17 You are given the distribution.

9	22	17	33	34	26	19	27	20	19	33
7	13	38	23	17	31	39	6	18	4	16

Draw an ordered stem-and-leaf plot to show the distribution.

Use the key $2|2 = 22$,

Stem	leaf
0	
1	
2	
3	
4	

[3]

For Examiner's Use

18 The list shows the marks scored by a group of learners in a test.

The marks are arranged in order of size starting with the least.

x	11	19	23	y	67

(a) The range is 64.

Find the value of x.

Answer (a) $x =$ .... [1]

(b) The mean is 27.

Find the value of y.

Answer (b) $y =$ .... [2]

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